

Lattices & Boolean Algebra:-

Introduction:-

R is a Relation on a non empty set A is a subset of $A \times A$.

If $(x, y) \in R$ then xRy

Equivalence Relation:

A relation R is said to be an equivalence relation if R is

- i) Reflexive $\{ a \leq a \}$ $aRa \forall a$
- ii) Symmetric $\{ aRb \& bRa \}$ If aRb then bRa
- iii) Transitive aRb, bRc then aRc

Partial ordering Relation:

A relation R is said to be partial ordering relation on A

- i) Reflexive (aRa)
- ii) Antisymmetric $(aRb \& bRa \Rightarrow a=b)$
- iii) Transitive $aRb \& bRc \Rightarrow aRc$

Partially ordered Set (or) a Poset:

If R is a Partial ordering on A then (A, R) is called Poset.

Eg: 1

$$R = \{ (1,1), (1,2), (1,3), (2,2), (2,3), (3,3) \}$$

i) $(1,1), (2,2), (3,3)$ is reflexive.

ii) $2 \leq 3$ but $3 \not\leq 2$ so ~~not~~^{no} symmetry.

iii) $2 \leq 2$ & $2 \leq 3$

$$\therefore 2 \leq 3.$$

Eg: ① $(\mathbb{N}, \leq)$ is a poset.

Poset:

A set together with a partial order relation R is called a partially ordered set (or)

Poset

Eg: Not a Poset

$\mathbb{Z}$, divisible $2|-2, -2|2 \Rightarrow \rightarrow$

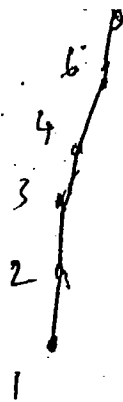
Hasse Diagram:

A partially ordered finite set $(P, \leq)$ can be graphically represented by a diagram called Hasse diagram of the poset.

Eg: ① Let $P = \{1, 2, 3, 4, 6, 12\}$

& Relation: $\leq$ $\Leftrightarrow a \leq b \Leftrightarrow b-a$ is non negative

$\therefore$ The Hasse Diagram $(P, \leq)$ is



it is a chain

Upper bounds & Least upper bounds:

In a poset $(P, \leq)$, Let A be the subset,

If $a \leq u$ for all $a \in A$
then u is called upper bound of A .

lub (supremum)

An upper bound which is less than all the other upper bounds called lub.

lower bounds & Greatest lower bounds:

In a poset $(P, \leq)$, Let A be the subset.

If $a \leq u$ for all $a \in A$
then u is called lower bound of A .

glb: (Infimum)

A lower bound which is greater than all the other lower bound called glb.

Note:

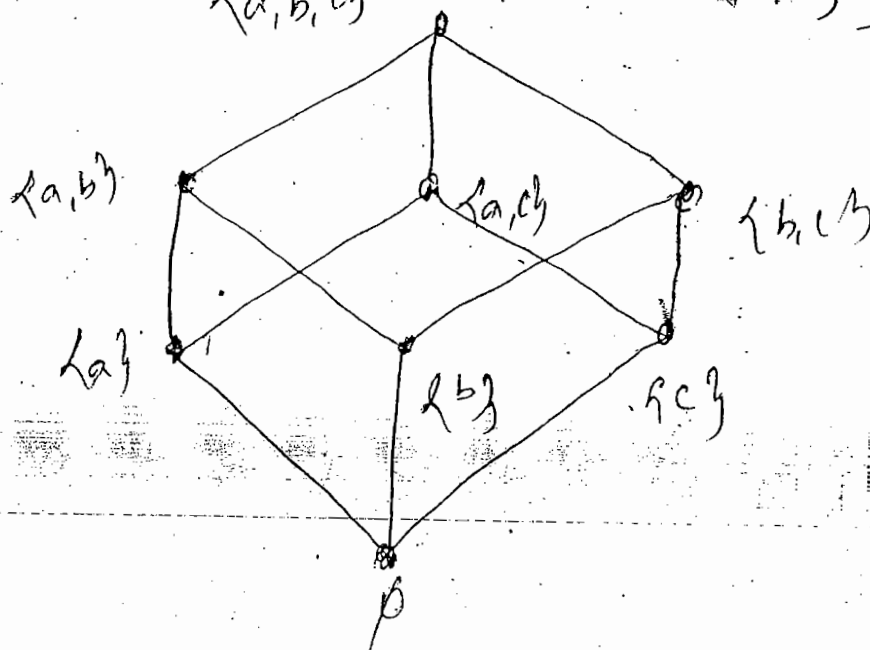
- ① If the lub of a & b exists
then it is denoted by $a \oplus b$ (or) $a \vee b$
(Join (or) Sum)
- ② If the glb of a & b exists
then it is denoted by $a \otimes b$ (or) $a \wedge b$
(meet (or) product)

Ques

- ① Draw the Hasse diagram which represents the partial ordering $\{(a, b) : a \leq b\}$ on the power set $P(A)$ where $A = \{a, b, c\}$.
& Find ub, lub, lb, glb, Maximal & minimal.

Sol:

$$P(A) = \{ \emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\} \}$$



u.b.:

The upper bound of the subset $(\{a\}, \{b\}, \{c\})$ is $\{a, b, c\}$

ii) Lub:

$\{a, b, c\}$

iii) Lb:

$\{a, b\}$ is not related to

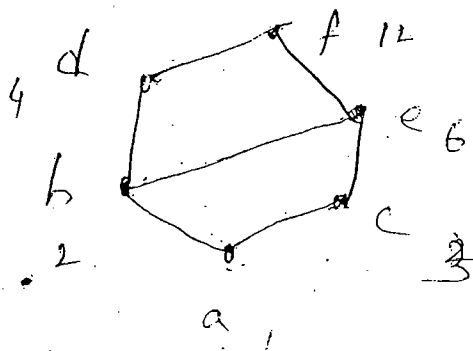
The lower bound of $(\{a, b\}, \{a, c\}, \{b, c\})$ is ϕ

iv) g.l.b : ϕ

$\{a, b\}, \{b\}, \{c\}$ are not lower bounds of the given subset.

②

Find the Lub & GLb for



$x \mid y$ $1, 2, 3, 4, 6, 12$
 $\leq$ be the relation "divisor of" on x

sol

i) Upper bounds of $\{a, b, c\}$ are e & f
 Since d is not an upperbound because it is not related to c

ii) LUB of $\{a, b, c\}$ is c

iii) lower bounds:

lb of $\{d, e\}$ are a & b

& Since c is not a lower bound,
Since c is not related to d

iv) glb:

glb of $\{d, e\}$ is b .

Lattice (L).

A lattice (L) is a poset ($L, \leq$)
in which every pair of elements $a, b \in L$
has a glb & lub.

NOTE:

① $\text{glb}\{a, b\} = \{a \star b\}$ { meet or product }
 $a \wedge b$ (or) $a \cdot b$

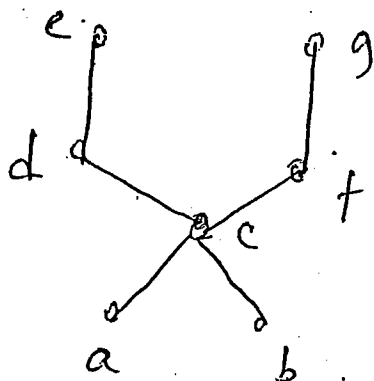
② $\text{lub}\{a, b\} = a \oplus b$ { Join (or) Sum }
 $a \vee b$ (or) $a + b$

③ A totally ordered set is trivially
a lattice.

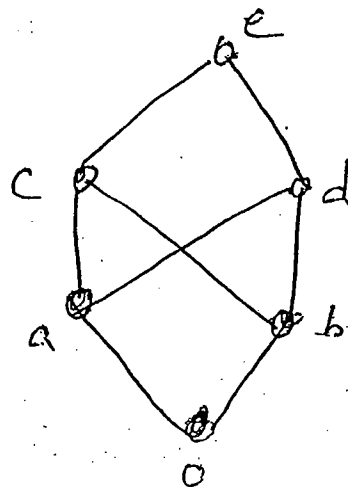
④ but All partially ordered sets are
not lattices.

Ex: for not a lattice

①



②



① $a \vee b$ exists ~~if~~ ^{each} if c

② $a \wedge b$ does not exist _{g.l.b}

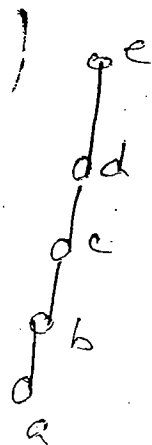
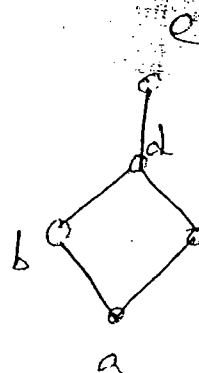
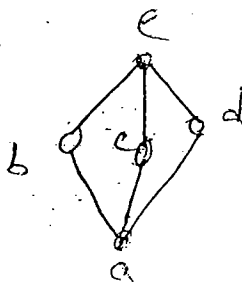
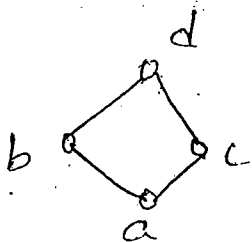
③ $d \wedge f = c$ but $d \vee f$ does not exist.

① $a \vee b$ does not exist

② $a \wedge b$ exists

For a lattice

①



Since every pair of elements of this poset has both LUB & a g.l.b.

⑧ $a = a^2$ is not true $\forall$ +ve Integers. (not Reflexive)

⑨ for Symmetric:

If aRb then bRa

If aRb mean $a = b^2$

does not imply $b = a^2$ $\forall$ +ve Integers.

$$25 = 5^2 \text{ but } 5 \neq 25^2$$

$\therefore$ NO Symmetry

⑩ anti Symmetry

If aRb i.e. $a = b^2$

& bRa i.e. $b = a^2$

then $a = b$

$$\text{i.e. } a = a^4$$

$$a^4 - a = 0$$

$$a(a^3 - 1) = 0 \quad (a \neq 0)$$

$$a^3 = 1$$

$$a = 1$$

$$\therefore a = b = 1$$

$\therefore R$ is anti symmetric.

$a \oplus b$

$a \star b$

$a \vee b$

$a \wedge b$

Sum

Meet

Lub

Glb

Let N be the set of all Nols.

Define a relation $\leq$ on N as follows
for $m, n \in N$, $m \leq n$ iff m is a divisor of
i.e. $n = km$.

i) $n \leq n$ for $n \in N$

ii) ~~m/n~~ m/n & n/m
i.e. $m = ln$ & $n = km$
 $n = lkm$

So $n = m$ is possible only when
 $kl = 1$

iii) $\leq$ is "Anti Symmetric"
i.e., $l=1$ & $k=1$

n divides m	$n = km$
m divides n	$m = ls$

$\therefore n$ divides s $\therefore n = kls$
 $\therefore n | s$

$(N, \leq)$ is a partial order relation.

2) N be the set of all the integers.

Let $\leq$ be the relation

Set $m \leq n$ iff $n-m$ is a non-negative integer.
is a poset.

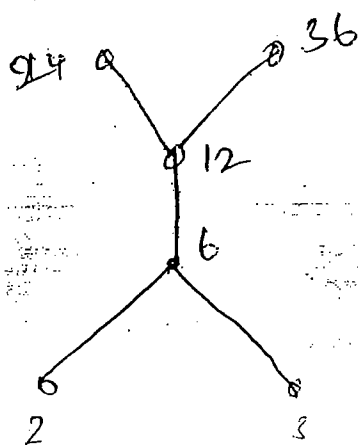
Note:

For a totally Ordered Set $(P, \leq)$
The Hasse diagram consists of circles
one below the other.

① Draw the Hasse diagram of $(\mathcal{N}, \leq)$

Where $\mathcal{N} = \{2, 3, 6, 12, 24, 36\}$

$\leq$ be the relation s.t if x div



② Determine whether the Relation R on
the set of all integers is reflexive,
symmetric, antisymmetric & / or transitive,
where aRb if & only if

i) $a \neq b$ ii) $ab \geq 0$ iii) $ab \geq 1$

iv) a is a multiple of b

v) $a \equiv b \pmod{7}$ vi) $|a-b| \geq 1$

vii) $a \geq b^2$ viii) $a \geq b^2$

Sol

- i) $a \neq a$ R is not reflexive
is not true
- ii) $a \neq b \Rightarrow b \neq a$ $\therefore R$ is Symmetric
- iii) $a \neq b$ & $b \neq c$ $\therefore R$ is not transitive
does not imply $a \neq c$

② $a^2 \geq 0$ $\therefore R$ is reflexive.

$ab \geq 0 \Rightarrow ba \geq 0$ R is Symmetric.

Consider $(2, 0)$ & $(0, -3) \in R$

But $(2, -3) \notin R$

Since $2(-3) < 0$

$\therefore R$ is not transitive

③ $a^2 \geq 1$ need not be true.
Since a may be zero. } not Reflexive

$ab \geq 1 \Rightarrow ba \geq 1$ $\therefore R$ is Symmetric

$ab \geq 1$ & $bc \geq 1 \Rightarrow a, b, c > 0$ (or) < 0

If all of $a, b, c > 0$,

$\therefore ac \geq 1$ (R is transitive)

④ i) a is a multiple of a .

R is Reflexive.

ii) If a is a multiple of b
 b is not a multiple of a

General. a is a multiple of b then

$\therefore R$ is anti symmetric

iii) When a is a multiple of b
& b is a multiple of c .
 $\therefore a$ is a multiple of c .

$\therefore R$ is Transitive.

$\therefore R$ is Reflexive, Anti-symmetric
& Transitive.

v) $(a-a)$ is a multiple of 7.

$\therefore R$ is reflexive

When $(a-b)$ is a multiple of 7
 $(b-a)$ is also a multiple of 7.
 $\therefore R$ is Symmetric.

When $a-b$ & $b-c$ are multiples of 7.
 $\therefore (a-b) + (b-c) = (a-c)$ is also a multiple of 7.

$\therefore R$ is transitive.

$\therefore R$ is Reflexive, Symmetric

& Transitive.

vi) $(a-a) \neq 0$

R is not reflexive.

$(a+b) \neq 1 \Rightarrow (b-a) = 1$

$\therefore R$ is Symmetric

$(a-b) \neq 1 \Rightarrow (b-a) = -1$

$$|b-c| = 1 \Rightarrow b-c = 1 \text{ or } -1$$

$$\textcircled{1} + \textcircled{2} \quad a-c = \pm 2 \text{ (or) } 0$$

$$\text{i.e. } |a-c| \neq 1$$

$\therefore R$ is Symmetric only.

vii) $a = a^2$ is not true for all integers.

$\therefore R$ is not Reflexive.

$$a = b^2 \text{ \& } b = a^2 \text{ for } a = b = 0 \text{ (or) } 1$$

$\therefore R$ is anti symmetric

$$a = b^2 \text{ \& } b = c^2 \text{ does not imply } a = c^2$$

$\therefore R$ is not Transitive.

viii)

$$a \geq a^2 \text{ is not true for all integers}$$

$\therefore R$ is not Reflexive.

$$a \geq b^2 \text{ \& } b \geq a^2 \text{ imply that } a = b$$

$\therefore R$ is anti symmetric.

$$\text{When } a \geq b^2 \text{ \& } b \geq c^2, a \geq c^2$$

$\therefore R$ is transitive

$\therefore R$ is anti symmetric \& transitive

Lattices

Def:

A Lattice is a partially ordered set $\langle L, \leq \rangle$ in which every pair of elements $a, b \in L$ has a glb & lub.

Note:

If $\langle S, \leq \rangle$ is a poset then $\langle S, \geq \rangle$ is also a poset.
& They are called duals of each other.

Properties of Lattices:

Idempotent
Commutative
Associative

$$a * a = a$$

$$a * b = b * a$$

$$(a * b) * c = a * (b * c)$$

$$i) \quad a * (a \oplus b) = a$$

$$a \oplus a = a$$

$$a \oplus b = b \oplus a$$

$$(a \oplus b) \oplus c = a \oplus (b \oplus c)$$

$$a \oplus (a * b) = a$$

(Absorption)

Proof?

L-4

(14)

for any $a \in L$

$$a \leq a$$

$$\& a \leq a \oplus b$$

$$\therefore a \leq a * a \oplus b \quad \text{--- (1)}$$

$$\& a * (a \oplus b) \leq a \quad \text{--- (2)}$$

from (1) & (2)

$$\therefore a * a \oplus b = a.$$

Theorem:

Let $(L, \leq)$ be a lattice in which $*$ and $\oplus$ denote the operations of meet & join respectively. For any $a, b \in L$,

$$a \leq b \iff a * b = a \iff a \oplus b = b$$

Proof:

First we will prove

$$\text{Let } a \leq b$$

$$a \leq b \iff a * b = a$$

$$\& \text{w.k.t } a \leq a$$

$$\therefore a \leq a * b \quad \text{--- (1)}$$

from

defn of $a * b$,

$$a * b \leq a$$

Note:

By def of $a * b$
 $a * b \leq a$

$a \oplus b$ (Lub)

$$a \oplus b \geq a$$

$$a * b = a.$$

$$\therefore a \leq b \Rightarrow a * b = a \quad \text{--- (1)}$$

Conversely, Let $a * b = a$

to prove: $a \leq b.$

$a * b = a$ is possible only when

$$a \leq b. \quad \text{--- (2)}$$

$\therefore$ from (1) & (2)

$$\therefore a \leq b \Leftrightarrow a * b = a$$

To prove: $a \leq b \Leftrightarrow a \oplus b = b.$

from $a * b = a$

$$b \oplus (a * b) = b \oplus a \\ = a \oplus b \quad \text{--- (3)}$$

by absorption law,

$$b \oplus (a * b) = b \quad \text{--- (4)}$$

from (3) & (4) $a \oplus b = b \quad \text{--- (5)}$

conversely Let $a \oplus b = b$

This is possible only when.
 --- (6)

Property:

Theorem:

Let $(L, \leq)$ be a lattice. For any $a, b \in L$
the following properties holds.

If $b \leq c$ then i) $a * b \leq a * c$

ii) $a \oplus b \leq a \oplus c$

Proof

w.k.T, $b \leq c$

$$\Leftrightarrow b * c = b$$

To Prove: $a * b \leq a * c$

i.e. $(a * b) * (a * c) = a * b$

LHS: $(a * b) * (a * c)$

$$= a * (b * a) * c$$

$$= a * (a * b) * c$$

$$= (a * a) * (b * c)$$

$$= a * (b * c)$$

$$= a * b = \text{RHS}$$

H.p

w.k.T $b \leq c$

$$\Leftrightarrow b \oplus c = c$$

To Prove:

$$a \oplus b \leq a \oplus c$$

i.e. $(a \oplus b) \oplus (a \oplus c) = a \oplus c$

LHS
 $(a \oplus b) \oplus (a \oplus c)$

$$= (a \oplus a) \oplus (b \oplus c)$$

$$= a \oplus (b \oplus c)$$

$$= a \oplus c = \text{RHS}$$

Note:

i) $a \leq b \wedge a \leq c \Rightarrow a \leq b * c$

ii) $a \leq b \wedge a \leq c \Rightarrow a \leq b \oplus c$

iii) $a \geq b \wedge a \geq c \Rightarrow a \geq b * c$

iv) $a \geq b \wedge a \geq c \Rightarrow a \geq b \oplus c$

Distributive Inequalities:-

Theorem - 3

Let $(L, \leq)$ be a lattice. For any $a, b, c \in L$ the following inequalities holds.

$$a \oplus (b * c) \leq (a \oplus b) * (a \oplus c)$$
$$a * (b \oplus c) \geq (a * b) \oplus (a * c)$$

Proof: from def of lub,

$$a \leq a \oplus b \text{ \& } a \leq a \oplus c$$

Since $a \leq b$ \& $a \leq c \Rightarrow a \leq b * c$ from (ii)

$$a \leq (a \oplus b) * (a \oplus c) \quad \text{--- (i)}$$

Also, $b * c \leq b \leq a \oplus b$

$$b * c \leq c \leq a \oplus c$$

$$\therefore b * c \leq (a \oplus b) * (a \oplus c) \quad \text{from (i)}$$

(ii)

$\therefore$ from (iv)

$$a \oplus (b * c) \leq (a \oplus b) * (a \oplus c)$$

||| ^{by} we can prove $a * (b \oplus c) \geq (a * b) \oplus (a * c)$

...ines

// Let $(L, \leq)$ be a lattice. For any $a, b, c \in L$ the following inequalities holds.

i) $a \leq c \Leftrightarrow a \oplus (b * c) \leq (a \oplus b) * c$

ii) $a \geq c \Leftrightarrow a * (b \oplus c) \geq (a * b) \oplus c$

Proof:

If $a \leq c$ then $a \oplus c = c$

& from $a \oplus (b * c) \leq (a \oplus b) * (a \oplus c)$
 $\leq a \oplus b * c$

III ly we can prove

$a \geq c \Leftrightarrow a * b \oplus c \geq \cancel{(a \oplus b)} * \cancel{c}$
 $(a \oplus b) * c$

Theorem:

In a lattice $(L, \leq)$ s.t.

i) $(a * b) \oplus (c * d) \leq (a \oplus c) * (b \oplus d)$

ii) $(a * b) \oplus (b * c) \oplus (c * a) \leq (a \oplus b) * (b \oplus c) * (c \oplus a)$
 $\forall a, b, c \in L$

Proof

Let $a, b, c \in L$

then $a * b \leq a \leq a \oplus b$ — ①

$a * b \leq a \leq c \oplus a$ — ②

$a * b \leq b \leq b \oplus c$ — ③

from ①, ② & ③

by
 III $a * b \leq (a \oplus b) * (b \oplus c) * (c \oplus a)$
 $b * c \leq (a \oplus b) * (b \oplus c) * (c \oplus a)$
 $c * a \leq (a \oplus b) * (b \oplus c) * (c \oplus a)$

$$\therefore (a * b) \oplus (b * c) \oplus (c * a) \leq (a \oplus b) * (b \oplus c) * (c \oplus a)$$

$$\therefore \text{Also } a \leq a \oplus c$$

$$b \leq b \oplus d$$

$$\therefore (a * b) \leq (a \oplus c) * (b \oplus d)$$

$$\& \quad c \leq a \oplus c$$

$$d \leq b \oplus d$$

$$\therefore c * d \leq (a \oplus c) * (b \oplus d)$$

$$\therefore (a * b) \oplus (c * d) \leq (a \oplus c) * (b \oplus d)$$

Note:

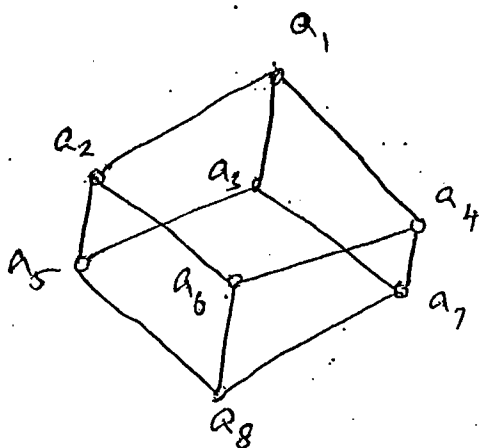
A lattice is an algebraic system $(L, *, \oplus)$ with two binary operations $*$ & $\oplus$ on L which are both

- ① Commutative
- ② associative & satisfy
- ③ absorption laws.

Sublattice:

Let $(L, *, \oplus)$ be a lattice & let $S \subseteq L$ be a subset of L . The algebra $(S, *, \oplus)$ is a sublattice of $(L, *, \oplus)$ iff S is closed under both operations $*$ & $\oplus$.

Eg: Let $(L, \leq)$ be a lattice in which $L = \{a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8\}$



$$S_1 = \{a_1, a_2, a_4, a_6\}$$

$$S_2 = \{a_2, a_5, a_7, a_8\}$$

$$S_3 = \{a_1, a_2, a_4, a_8\}$$

be the subsets.

$(L, \leq) \rightarrow$ a lattice

$(S_1, \leq) \rightarrow$ a lattice

$(S_2, \leq) \rightarrow$ a lattice

$(S_3, \leq) \rightarrow$ not a lattice

since, $a_2, a_4 \in S_3$

but $a_2 \vee a_4 = a_6 \notin S_3$.

NOTE :

Let $(L, *, \oplus)$ be a lattice & $S \subseteq L$ be a finite subset of L where

$$S = \{a_1, a_2, \dots, a_n\}$$

$$\therefore \text{glb } S = \bigwedge_{i=1}^n a_i \text{ \& \; } \text{lub } S = \bigvee_{i=1}^n a_i$$

$$= a_1 * a_2 * \dots * a_n$$

$$= a_1 \oplus a_2 \oplus \dots \oplus a_n$$

Defn: Complete lattice:

A lattice L is called complete if each of its non empty subsets has a lub & glb.

Note:

① Every finite lattice must be complete. & It must have a least element & greatest element. If they exists they are called the bounds.

② The bounds are denoted by 0 & 1 respectively.

$0 \rightarrow$ lower bound if $0 \leq a \forall a \in L$
 $1 \rightarrow$ upper bound if $a \leq 1 \forall a \in L$

Bounded lattice:

A lattice which has both elements 0 & 1 is called a bounded lattice.

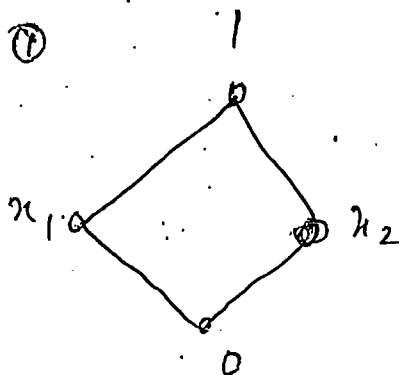
Eg: Let $\langle L, *, \oplus \rangle$ be a lattice.

With $L = \{a_1, a_2, \dots, a_n\}$,

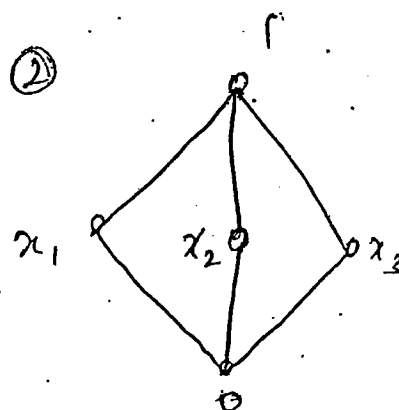
$$\bigstar_{i=1}^n a_i = 0 \quad \& \quad \bigoplus_{i=1}^n a_i = 1$$

Where 0 & 1 are bounds of a lattice

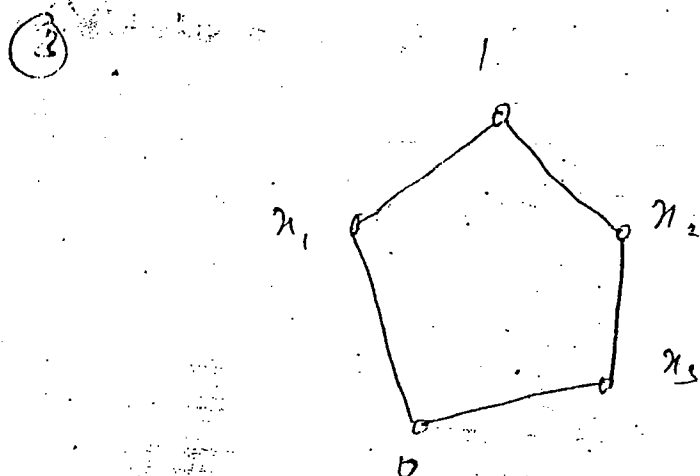
eg:



Complement of x_1 is x_2



Complements of x_1 are x_2, x_3
Complements of x_2 are x_1, x_3



Complements of x_1 are x_2 & x_3

Complemented lattice:

A lattice $\langle L, *, \oplus, 0, 1 \rangle$ is said to be complemented lattice if every element of L has at least one complement.

Distributive lattice:

A lattice $(L, *, \oplus)$ is called a distributive lattice if for any $a, b, c \in L$

$$a * (b \oplus c) = (a * b) \oplus (a * c)$$

$$a \oplus (b * c) = (a \oplus b) * (a \oplus c)$$

i.e., $*$, $\oplus$ are distributive over each

Note:

① A bounded lattice is always denoted by $\langle L, *, \oplus, 0, 1 \rangle$

② It should satisfy the following identities.

for any $a \in L$

$$a \oplus 0 = a$$

$$a * 1 = a$$

$$a * 1 = 1$$

$$a \oplus 0 = 0$$

Def: Complement of an element in a Lattice:

In a bounded lattice $\langle L, *, \oplus, 0, 1 \rangle$ an element $b \in L$ is called a Complement of an element $a \in L$ if

$$a * b = 0 \quad \& \quad a \oplus b = 1$$

Note:

① Complement ^{possess} ~~to~~ Symmetry, ~~in $a \oplus b$~~

② If a is complement of b
then b is complement of a

③ for any $a \in L$ May or May not have a Complement.

④ Complement is not unique.

⑤ $0 * 1 = 0$ & $0 \oplus 1 = 1$

$\therefore 0$ & 1 are complements of each other.

Thm:

Every chain is a distributive lattice.

Proof Let $(L, \leq)$ be a chain.

for $a, b, c \in L$, consider,

Case (i) $a \leq b$ (or) $a \leq c$

Case (ii) $a \geq b$ & $a \geq c$

for (i) $a * (b \oplus c) = a$ & $(a * b) \oplus (a * c) = a$ — (1)

for (ii) $a * (b \oplus c) = b \oplus c$ & $(a * b) \oplus (a * c) = b \oplus c$ — (2)

$\therefore$ from (1) & (2)

$$a * (b \oplus c) = (a * b) \oplus (a * c)$$

hence we can prove

$$a \oplus (b * c) = (a \oplus b) * a \oplus c$$

$\therefore$ Every chain is a lattice.

② A lattice is said to be modular

if $a \leq c \Rightarrow a \oplus b * c = a \oplus b * c$

S.T every distributive lattice is modular, but not conversely,

Proof

Since distributive lattice

$$a \oplus (b \wedge c) = (a \oplus b) \wedge (a \oplus c)$$

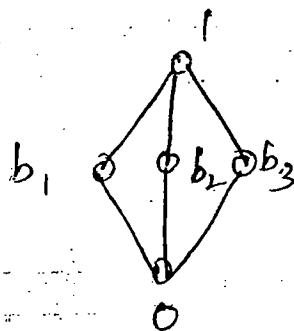
Since $a \leq c$

$$\therefore a \oplus c = c$$

$$\therefore a \oplus (b \wedge c) = (a \oplus b) \wedge c$$

Conversely,

Let us consider the ex.



$$b_1 \oplus (b_2 \wedge b_3) = b_1 \wedge 1 = b_1 \quad \text{--- (1)}$$

$$(b_1 \wedge b_2) \oplus (b_1 \wedge b_3) = 0 \oplus 0 = 0 \quad \text{--- (2)}$$

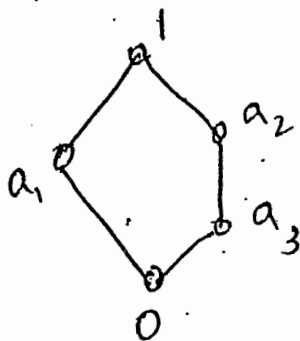
from (1) & (2)

It is not

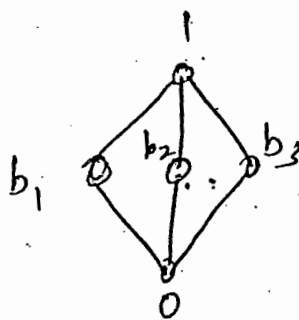
distributive.

Sum

Show that the lattices given by are



①



not distributive

Sol

$$a_1 * (a_1 \oplus a_2) = a_3 * 1 = a_3$$

$$(a_3 * a_1) \oplus (a_3 * a_2) = 0 \oplus a_3 = a_3$$

$$\& a_1 * (a_2 \oplus a_3) = 0$$

$$(a_1 * a_2) \oplus (a_1 * a_3) = 0 \oplus 0 = 0$$

$$\text{but } a_2 * (a_1 \oplus a_3) = a_2 * 1 = a_2$$

$$(a_2 * a_1) \oplus (a_2 * a_3) = 0 \oplus a_3 = a_3$$

$\therefore$ ① is not distributive.

$$\text{from ② } b_1 * (b_2 \oplus b_3) = b_1 * 1 = b_1$$

$$(b_1 * b_2) \oplus (b_1 * b_3) = 0 \oplus 0 = 0$$

$\therefore$ ② is not distributive

Note: If $a \leq c$ then $a * c = a$ and $a \oplus c = c$

no index

Theorem: -

Let $(L, *, \oplus)$ be a distributive lattice

For any $a, b, c \in L$

$$(a * b = a * c) \wedge (a \oplus b \geq a \oplus c)$$

$$\Rightarrow b \geq c$$

Proof:

$$(a * b) \oplus c = (a * c) \oplus c = c$$

$$\begin{aligned}(a * b) \oplus c &= (a \oplus c) * (b \oplus c) = (a \oplus b) * (b \oplus c) \\ &= \cancel{b} * (a * c) = \cancel{b} * (a * b) \\ &= b\end{aligned}$$

$$\therefore b = c$$

Aliter:

$$\begin{aligned}b &= b * (b \oplus a) \\ &= b * (a \oplus c) \\ &= (b * a) \oplus (b * c) \\ &= (a * c) \oplus (b * c) \\ &= (a \oplus b) * c \\ &= (a \oplus c) * c \\ &= c\end{aligned}$$

Def:

A Boolean algebra is a Complemented distributive lattice.

It is denoted by $(B, *, \oplus, ', 0, 1)$

Properties of Boolean Algebra:

I For a lattice $(B, *, \oplus)$ the $\sim$ properties holds

① $a * a = a$

$a \oplus a = a$

② $a * b = b * a$

$a \oplus b = b \oplus a$

③ $(a * b) * c = a * (b * c)$

$(a \oplus b) \oplus c = a \oplus (b \oplus c)$

④ $a * (a \oplus b) = a$

$a \oplus (a * b) = a$

II For a distributive lattice $(B, *, \oplus)$ the $\sim$ properties holds:

① $a * (b \oplus c) = (a * b) \oplus (a * c)$

② $a \oplus (b * c) = (a \oplus b) * (a \oplus c)$

③ $(a * b) \oplus (b * c) \oplus (c * a) = (a \oplus b) * (b \oplus c) * (c \oplus a)$

④ $a * b = a * c \ \& \ a \oplus b = a \oplus c$
 $\Rightarrow b = c$

III For a Bounded lattice:

$\langle B, *, \oplus, 0, 1 \rangle$ is a bounded lattice in

which for $a \in B$,

① $0 \leq a \leq 1$

② $a * 0 = 0$

$a \oplus 1 = 1$

③ $a * 1 = a$

$a \oplus a = a$

IV for a uniquely complemented lattice

$$\langle B, *, \oplus, ', 0, 1 \rangle$$

for any $a \in B$ the complement $a' \in B$

Satisfies,

$$① \quad a * a' = 0$$

$$a \oplus a' = 1$$

$$② \quad 0' = 1$$

$$1' = 0$$

$$③ \quad (a * b)' = a' \oplus b'$$

$$(a \oplus b)' = a' * b'$$

↓ $\exists$ a partial order relation $\leq$ on B s.t.

$$① \quad a * b = \text{GLB} \{a, b\}$$

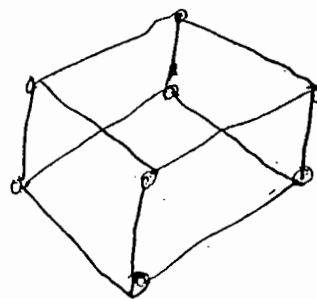
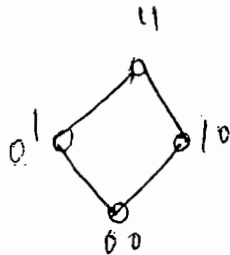
$$a \oplus b = \text{LUB} \{a, b\}$$

$$② \quad a \leq b \Leftrightarrow a * b = a$$

$$\Leftrightarrow a \oplus b = b$$

$$③ \quad a \leq b \Leftrightarrow a * b' = 0 \Leftrightarrow b' \leq a' \Leftrightarrow a' \oplus b = 1$$

Bg:



Note

$$* \rightarrow \cdot \quad \text{i.e. } \wedge$$

$$\oplus \rightarrow \vee \quad \text{i.e. } \vee$$

$$(\text{Complement}) ' \rightarrow \text{Negation i.e. } \neg$$

$$0 \rightarrow F$$

$$1 \rightarrow T$$

Sub algebra:

Let $\langle B, *, \oplus, ', 0, 1 \rangle$ be a Boolean Algebra
& $S \subseteq B$ if S contains 0 & 1 and is
closed under $*$, $\oplus$ & $'$, then

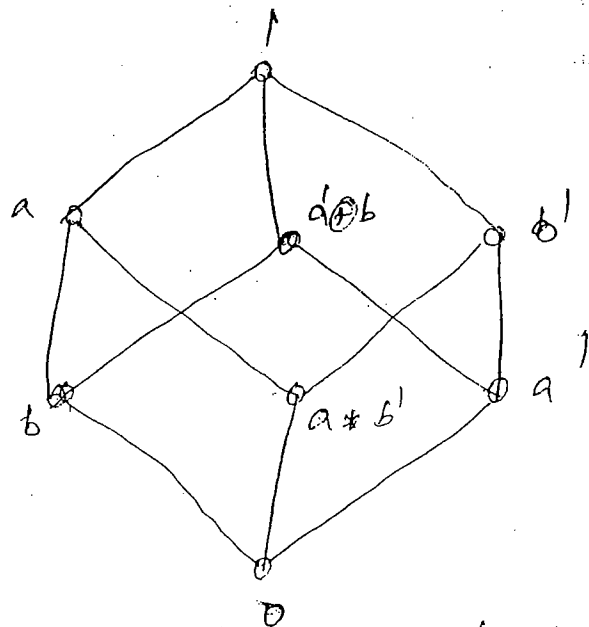
$\langle S, *, \oplus, ', 0, 1 \rangle$ is called a Sub boolean
algebra.

Note

$\{*, '\}$ (or) $\{\oplus, '\}$ are Functionally
complete in a Boolean Algebra.

Since $(a \oplus b)' = a' * b'$
 $1 = (a * a')'$ & $0 = a * a'$

Eg:



In this Boolean
Algebra.

$S_1 = \{a, a', 0, 1\}$, $S_2 = \{a \oplus b, a * b', 0, 1\}$
are sub-boolean algebras

$S_3 = \{a * b', b', a, 0, 1\}$ is not a sub-boolean
algebra.

Properties:

$$B_1: \left. \begin{array}{l} a+0 = a \\ a \cdot 1 = a \end{array} \right\} \text{Identity laws}$$

$$B_2: \left. \begin{array}{l} a+b = b+a \\ a \cdot b = b \cdot a \end{array} \right\} \text{Commutative laws}$$

$$B_3: \left. \begin{array}{l} (a+b)+c = a+(b+c) \\ (a \cdot b) \cdot c = a \cdot (b \cdot c) \end{array} \right\} \text{Associative laws}$$

$$B_4: \left. \begin{array}{l} a+(b \cdot c) = (a+b) \cdot (a+c) \\ a \cdot (b+c) = (a \cdot b) + (a \cdot c) \end{array} \right\} \text{Distributive laws}$$

$$B_5: \left. \begin{array}{l} a+a' = 1 \\ a \cdot a' = 0 \end{array} \right\} \text{Complement laws}$$

Note:

$$0+0 = 0$$

$$0+1 = 1 = 1+0 = 1$$

$$0 \cdot 0 = 0 \cdot 1 = 1 \cdot 0 = 0$$

$$1 \cdot 1 = 1$$

$$0' = 1 \quad \& \quad 1' = 0$$

Additional Properties of Boolean Algebra:

i) Idempotent laws:

$$a+a = a$$

$$\& \quad a \cdot a = a \quad \forall a \in B$$

ii) Dominance laws

$$a+1 = 1$$

$$\& \quad a \cdot 0 = 0 \quad \text{for all } a \in B$$

iii) Absorption laws

$$a \cdot (a+b) = a$$

$$\& \quad a+(a \cdot b) = a$$

$$\forall a, b \in B$$

iv) De Morgan's laws

① In a Boolean algebra, If $a+b=1$ & $a \cdot b=0$,
S.t $b=a'$

i.e, The complement of every element is
unique.

Proof:

$$b = b \cdot 1$$

$$= b \cdot (a + a')$$

$$= b \cdot a + b \cdot a'$$

$$= a \cdot b + a' \cdot b = 0 + a' \cdot b = a \cdot a' + a' \cdot b$$

$$= a \cdot a' + a' \cdot b$$

$$= a' \cdot (a + b)$$

$$b = a'$$

② In a Boolean algebra the following statements
are equivalent.

$$\textcircled{1} a+b=b \quad \textcircled{2} a \cdot b=a \quad \textcircled{3} a'+b=1 \quad \textcircled{4} a \cdot b'=0$$

Proof

Let $\textcircled{1}$ be true.

$$\begin{aligned} \text{then } a \cdot b &= a \cdot (a+b) \\ &= a \end{aligned}$$

$$\textcircled{1} \Rightarrow \textcircled{2}$$

Conversely

$$\text{Let } a \cdot b = a$$

$$\begin{aligned} a+b &= (a \cdot b) + b \\ &= b \end{aligned}$$

$$\textcircled{2} \Rightarrow \textcircled{1}$$

$$a' + b = a' + a + b \\ = (a + a') + b = 1 + b = 1$$

$$\textcircled{1} \Rightarrow \textcircled{3}$$

Conversely, let $a' + b = 1$

$$\therefore a + b = (a + b) \cdot 1 \\ = (a + b) \cdot (a' + b) \\ = a \cdot a' + b \\ = 0 + b \\ = b$$

$$\textcircled{3} \Rightarrow \textcircled{1}$$

$$\therefore \textcircled{1} \Leftrightarrow \textcircled{3}$$

Let $a' + b = 1$

$$(a' + b)' = 1' \\ (a')' \cdot b' = 0$$

$$a \cdot b' = 0$$

$$\therefore \textcircled{3} \Rightarrow \textcircled{4}$$

Conversely, let $a \cdot b' = 0$

$$a' + (b')' = 0'$$

$$a' + b = 1$$

$$\textcircled{4} \Rightarrow \textcircled{3}$$

$$\textcircled{3} \Leftrightarrow \textcircled{4}$$

2. Simplify the Boolean Expression.

$$(a' * b' * c) \oplus (a * b' * c) \oplus (a * b' * c')$$

Sol

$$(a' * b' * c) \oplus (a * b' * c) \oplus (a * b' * c')$$

$$= (a' * b' * c) \oplus (a * b') * (c \oplus c')$$

$$= (a' * b' * c) \oplus (a * b') * 1$$

$$= (a' * b' * c) \oplus (a * b')$$

$$= b' * ((a' * c) \oplus a)$$

$$= b' * \{(a' \oplus a) * (a \oplus c)\}$$

$$= b' * (a \oplus c)$$

$$= a * b' \oplus b' * c$$

(4) In any Boolean algebra p.t

$$a * b' \oplus a' * b = 0 \quad \text{iff} \quad a = b.$$

Proof:

Let $a = b$

$$\begin{aligned} (a * b') \oplus (a' * b) &= a * a' \oplus a' * a \\ &= 0 \oplus 0 = 0 \quad \text{--- (1)} \end{aligned}$$

Conversely,

$$(a * b') \oplus (a' * b) = 0$$

$$a \oplus (a * b') \oplus (a' * b) = a \oplus 0$$

$$a \oplus (a' * b) = a$$

$$(a \oplus a) * (a \oplus b) = a$$

$$1 * (a \oplus b) = a$$

$$a \oplus b = a \quad \text{--- ①}$$

|||ly

$$(a * b') \oplus (a' * b) \oplus b = 0 \oplus b$$

$$(a * b') \oplus b = b$$

$$(a \oplus b) * (b' \oplus b) = b$$

$$a \oplus b = b \quad \text{--- ②}$$

$$\text{from ① \& ②} \quad a = b$$

⑤. In any Boolean Algebra s.t.

$$(a \oplus b') * (b \oplus c') * (c \oplus a') = (a' \oplus b) * (b' \oplus c) * (c' \oplus a)$$

Proof

$$\underline{\text{LHS}} : (a \oplus b') * (b \oplus c') * (c \oplus a')$$

$$= ((a \oplus b') \oplus (c * c')) * ((b \oplus c') \oplus (a * a)) * ((c \oplus a') \oplus (b * b'))$$

$$= ((a \oplus b') \oplus c) * ((a \oplus b') \oplus c') * ((b \oplus c') \oplus a)$$

$$* ((b \oplus c') \oplus a') * ((c \oplus a') \oplus b)$$

$$* ((c \oplus a') \oplus b) * ((c \oplus a') \oplus b')$$

$$= ((a' \oplus b) \oplus c') * ((a' \oplus b) \oplus c) * ((c' \oplus a) \oplus b)$$

$$= \{ (a' \oplus b) \oplus (c' \oplus c) \} * \{ (c' \oplus a) \oplus (b * b') \} \\ * (b' \oplus c) \oplus (a * a')$$

$$= (a' \oplus b) + (b' \oplus c) + (c' \oplus a)$$

$$= \text{RHS.}$$

(6) In any Boolean Algebra, p.t.

$$\text{i) } x \oplus (w * y) \oplus (u * z) = (x \oplus u \oplus w) * (x \oplus v \oplus w) * (x \oplus u \oplus y) \\ * (x \oplus v \oplus y) * (x \oplus w \oplus z) * (x \oplus y \oplus z)$$

$$\text{ii) } (a * b) \oplus (a * b) * c \oplus (a' + b) \oplus a + b' * c = b \oplus a * c$$

Proof

RHS

$$\text{i) } = (x \oplus u \oplus w) * (x \oplus u \oplus y) * (x \oplus v \oplus w) * (x \oplus v \oplus y) \\ * (x \oplus w \oplus z) * (x \oplus y \oplus z)$$

$$= (x \oplus u) \oplus (w * y) * (x \oplus v) \oplus (w * y) * (x \oplus z) \oplus (w * y)$$

$$= \{ (x \oplus (w * y)) \oplus (u * v) \} * x \oplus (w * y) \oplus z$$

$$= (x \oplus (w * y)) \oplus (u * v * z) = \text{LHS}$$

$$\text{ii) } \text{LHS} = (a * b) \oplus (a' * b) \oplus (a * b * c) \oplus (a * b' * c)$$

$$= (a \oplus a') * b + (b \oplus b') * (a * c)$$

$$= b \oplus (a * c)$$

$$= \text{RHS.}$$

⑦. Simplify using Boolean Algebra.

~~i) $(x \oplus y \oplus x \oplus y)$~~

i) $(x \oplus y \oplus (x \oplus y)) \oplus (x \oplus z)$

ii) $x \oplus \{ y \oplus z \oplus ((x \oplus y) \oplus (x \oplus z))' \}$

iii) $(x \oplus y)' \oplus z \oplus (x' \oplus y) \oplus z'$

iii)
 $(x \oplus y)' \oplus z$
 $\oplus (x' \oplus y) \oplus z'$

$= (x \oplus y)' \oplus z$
 $\oplus ((x \oplus y) \oplus z)'$
 $= 1$

Proof

i) $(x \oplus y \oplus (x \oplus y)) \oplus (x \oplus z)$

$= (x \oplus y) \oplus (x \oplus z)$

~~$= (x \oplus x) \oplus (x \oplus z) \oplus (y \oplus x) \oplus (y \oplus z)$~~

~~$= (x \oplus (x \oplus z)) \oplus (y \oplus (y \oplus z))$~~

$= (x \oplus (x \oplus z)) \oplus (y \oplus (y \oplus z))$

$= x \oplus (y \oplus x) \oplus (y \oplus z)$

$= x \oplus (y \oplus z)$

ii) $x \oplus \{ y \oplus z \oplus ((x \oplus y) \oplus (x \oplus z))' \}$

$= x \oplus \{ y \oplus z \oplus ((x' \oplus y') \oplus (x' \oplus z')) \}$

$= x \oplus \{ y \oplus z \oplus (x' \oplus (x' \oplus z')) \oplus y' \oplus (x' \oplus z') \}$

$= x \oplus \{ y \oplus z \oplus (x' \oplus ((y' \oplus x') \oplus (y \oplus z))) \}$

$= x \oplus \{ y \oplus z \oplus (x' \oplus (y' \oplus z')) \}$

$= x \oplus \{ y \oplus (z \oplus x') \oplus z \oplus (y' \oplus z') \}$

$= x \oplus \{ y \oplus (z \oplus x') \}$

Boolean functions:

Consider a set of n variables $x_1, x_2, \dots, x_n$ & operator symbols $\wedge, \vee, \neg$ with the help of these symbols & variables we form strings according to certain rule.

Def: A Boolean Expressions, form or formula in n variables $x_1, x_2, \dots, x_n$ in any finite string of symbols formed in the following manner

- ① 0 & 1 are Boolean expressions.
- ② $x_1, x_2, \dots, x_n$ are "
- ③ If α_1 & α_2 are Boolean expressions $(\alpha_1) \wedge (\alpha_2)$ & $(\alpha_1) \vee (\alpha_2)$ are also Boolean Expressions.
- ④ If α is a Boolean expression then α' is also a "
- ⑤ No strings of symbols except those formed in accordance with rules 1 to 4 are Boolean Expressions.

Def: Two Boolean forms $\alpha(x_1, x_2, \dots, x_n)$ & $\beta(x_1, x_2, \dots, x_n)$ are called equivalent if one can be obtained from the other by a finite no. of applications of the identities.

Minterms in Boolean Algebra:

A Boolean form in n variables

$x_1, x_2, \dots, x_n$ Consisting of the product of n terms

$$\text{S.t. } x_1^{a_1} \& x_2^{a_2} \& \dots \& x_n^{a_n} = \prod_{i=1}^n x_i^{a_i}$$

in which a_i is either 0 (or) 1

x_i^0 stands for x_i

& x_i^1 stands for x_i' for $i = 1, 2, \dots, n$ is

Called minterm (or) complete product or a fundamental product.

Note:

① If minterms may come

Consider $x_i \rightarrow 1 \quad \forall i = 1, 2, \dots, n$

$x_i' \rightarrow 0 \quad \text{for } i = 1, 2, \dots, n$

② If Max terms may come

Consider $x_i' \rightarrow 1 \quad \text{for } i = 1, 2, \dots, n$

$x_i \rightarrow 0 \quad \text{for } i = 1, 2, \dots, n$

③ for Max terms (CNF) for minterm (DNF)

$$000 = 0$$

$$001 = 1$$

$$010 = 2$$

$$011 = 3$$

$$100 = 4$$

$$101 = 5$$

$$110 = 6$$

$$111 = 7$$

$$111 = 7$$

$$110 = 6$$

$$101 = 5$$

$$100 = 4$$

$$011 = 3$$

$$010 = 2$$

$$001 = 1$$

$$000 = 0$$

known as Sum of minterms (SOP) is known as Sum of minterms (SOP)

⑤ Product of Sums of Canonical form is known as Product of Maxterms (POCF)

Sums

① Write the Boolean expression in an equivalent Sum of Products Canonical form in three variables.

i) $x_1 \oplus x_2$ ii) $x_1 \oplus x_2$ iii) $(x_1 \oplus x_2)' \oplus x_3$

PDNF:

$$\begin{aligned} x_1 \oplus x_2 &= (x_1 \oplus x_2) * (x_3 \oplus x_3') \\ &= (x_1 \oplus x_2 \oplus x_3) \oplus (x_1 \oplus x_2 \oplus x_3') \\ &= m_7 \oplus m_6 = \oplus 6, 7 \end{aligned}$$

PCNF

$$\begin{aligned} x_1 \oplus x_2 &= (x_1 \oplus (x_2 \oplus x_2')) * (x_2 \oplus (x_1 \oplus x_1')) \\ &= (x_1 \oplus x_2) * (x_1 \oplus x_2') * (x_2 \oplus x_1) * (x_2 \oplus x_1') \\ &= (x_1 \oplus x_2) * (x_1 \oplus x_2') * (x_2 \oplus x_1') \\ &= (x_1 \oplus x_2) \oplus (x_3 \oplus x_3') * (x_2 \oplus x_2') \oplus (x_3 \oplus x_3') \\ &= (x_1 \oplus x_2 \oplus x_3) * (x_1 \oplus x_2 \oplus x_3') * (x_1 \oplus x_2' \oplus x_3') \\ &\quad * (x_1 \oplus x_2' \oplus x_3) * (x_1' \oplus x_2 \oplus x_3) \end{aligned}$$

$$\geq M_{an_0} \& M_{an_1} \& M_{an_2} \& M_{an_3} \& M_{an_4} \& M_{an_5}$$

$$\geq \& 0, 1, 2, 3, 4, 5$$

Aliter

$$PDF \text{ of } x_1 * x_2 = (x_1 \& x_2 \& x_3) \oplus (x_1 \& x_2 \& x_3')$$

$$\begin{aligned} 7S &= (x_1 \& x_2' \& x_3) \oplus (x_1' \& x_2 \& x_3) \oplus (x_1' \& x_2' \& x_3) \\ &\oplus (x_1' \& x_2 \& x_3') \oplus (x_1 \& x_2' \& x_3') \\ &\oplus (x_1' \& x_2' \& x_3') \end{aligned}$$

$$\begin{aligned} 77S &= (x_1' \oplus x_2 \oplus x_3') * (x_1 \oplus x_2' \oplus x_3') * (x_1 \oplus x_2 \oplus x_3) \\ &* (x_1 \oplus x_2' \oplus x_3) * (x_1' \oplus x_2 \oplus x_3) \\ &* (x_1 \oplus x_2 \oplus x_3) \end{aligned}$$

Aliter

Truth table - Method:

x_1	x_2	x_3	$x_1 * x_2$
1	1	1	1
1	1	0	1
1	0	1	0
1	0	0	0
0	1	1	0
0	1	0	0
0	0	1	0
0	0	0	0

For

~~Sums of products~~
~~Product~~ ~~Sums~~
~~Sum of products~~ (P.P)
Product of sums (P.S)

Find 0's in the last resultant column

PCNF of $x_1 \& x_2$

$$\begin{aligned} &= (x_1 \oplus x_2 \oplus x_3) * (x_1 \oplus x_2 \oplus x_3') \\ &* (x_1 \oplus x_2' \oplus x_3) * \\ &(x_1 \oplus x_2' \oplus x_3') * (x_1' \oplus x_2 \oplus x_3') \\ &* (x_1' \oplus x_2 \oplus x_3') \end{aligned}$$

PDNF (Sum of the products) (41)

Find 1's in the last resultant column

∴ PDNF of $x_1 \oplus x_2$

$$= (x_1 \oplus x_2 \oplus x_3) \oplus (x_1 \oplus x_2 \oplus x_3')$$

Sums: 2

② $x_1 \oplus x_2$ Find Sum of Products Canonical form
Product of Sums Canonical form

③ $(x_1 \oplus x_2)' \oplus x_3$

Ans.

$$\begin{aligned} (x_1 \oplus x_2)' \oplus x_3 &= (x_1' \oplus x_2') \oplus x_3 \\ &= \min_1 \end{aligned}$$

$$\begin{aligned} \text{Ans: } & (x_1 \oplus x_2 \oplus x_3) \oplus (x_1 \oplus x_2 \oplus x_3') \\ & \oplus (x_1 \oplus x_2' \oplus x_3) \oplus (x_1 \oplus x_2' \oplus x_3') \\ & \oplus (x_1' \oplus x_2 \oplus x_3) \oplus (x_1' \oplus x_2 \oplus x_3') \\ & = \min_1 \oplus \min_6 \oplus \min_5 \oplus \min_3 \oplus \min_7 \\ & = \oplus 2, 3, 4, 5, 6, 7 \end{aligned}$$

$$\begin{aligned} \text{Ans: } & (x_1 \oplus x_2 \oplus x_3) \oplus (x_1 \oplus x_2 \oplus x_3') \\ & = \max_0 \oplus \max_1 \\ & = x_1 \oplus x_2 \end{aligned}$$

$$\begin{aligned} & (x_1' \oplus x_2' \oplus x_3' \oplus x_4') \oplus (x_1' \oplus x_2' \oplus x_3' \oplus x_4) \oplus (x_1' \oplus x_2' \oplus x_3 \oplus x_4) \\ & \oplus (x_1' \oplus x_2' \oplus x_3 \oplus x_4') = x_1' \oplus x_2' \end{aligned}$$

$$\begin{aligned} \text{Ans: } & (x_1' \oplus x_2' \oplus x_3') \oplus (x_1' \oplus x_2' \oplus x_3) \\ & = x_1' \oplus x_2' \end{aligned}$$

5) Find the value of :

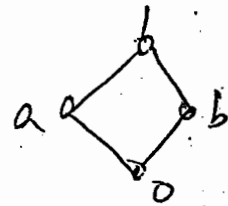
$$x_1 \oplus x_2 \oplus [(x_1 \oplus x_4) \oplus x_2' \oplus (x_3 \oplus x_1')]$$

for $x_1 = a$, $x_2 = 1$, $x_3 = b$, $x_4 = 1$ where $a, b, 1 \in$

the Boolean algebra is $\langle B, \oplus, \odot, ', 0, 1 \rangle$

So,

S.t.



$$(x_1 \oplus x_2 \oplus x_1 \oplus x_4) \oplus (x_1 \oplus x_2 \oplus x_2') \oplus (x_1 \oplus x_2 \oplus x_3 \oplus x_1')$$

$$= (x_1 \oplus x_2 \oplus x_4) \oplus 0 \oplus (x_1 \oplus x_2)$$

$$= (x_1 \oplus x_2 \oplus x_4) = a \oplus 1 \oplus 1 = a$$

Karnaugh Map:

$$x=1, y=1, z=1$$

① for 2 variables

x \ y	0	1
	0	1
0	$x'y'$	$x'y$
1	$x'y'$	$x'y$

② For 3 variables

x \ yz	00	01	11	10
	0	1	1	0
0	$x'y'z'$	$x'y'z$	$x'yz$	$x'yz'$
1	$xy'z'$	$xy'z$	xyz	xyz'

$$\begin{aligned} & \text{--- } x_2 = 1 \\ & \text{--- } x_2 = 2 \\ & \text{--- } x_2 = 4 \end{aligned}$$

$$\begin{aligned} & \text{--- } x_2 = 0 \\ & \text{--- } x_2 = 2 \\ & \text{--- } x_2 = 4 \end{aligned}$$

For 4 variables

w \ yz	00	01	11	10
	00	01	11	10
00	$w'x'y'z'$	$w'x'y'z$	$w'xyz'$	$w'xyz$
01	$w'xy'z'$	$w'xy'z$	$w'xyz'$	$w'xyz$
11	$wxy'z'$	$wxy'z$	$wxyz'$	$wxyz$
10	$wx'y'z'$	$wx'y'z$	$wx'yz'$	$wx'yz$

(43)

Find the Minimal Sum of Products Expressions for the function:

$$f(a, b, c) = ab'c' + abc' + abc + ab'c + a'b'c$$

(or)

$$a \oplus b' \oplus c' + a \oplus b \oplus c' + a \oplus b \oplus c + a \oplus b' \oplus c + a' \oplus b' \oplus c$$

Karnaugh table:

	bc	00	01	11	10
a					
0		0	1	0	0
1		1	1	1	1

Step: 1 ~~For~~

For each minterm place the value in the table.

Step: 2

Then loop the adjacent cells containing 1's

Step: 3

$$\begin{array}{r} 001 \\ 101 \\ \hline b'c \\ \hline \end{array} \quad \begin{array}{r} 100 \\ 101 \\ 111 \\ 110 \\ \hline a \\ \hline \end{array}$$

$$f(a, b, c) = a + b'c$$

② Minimise the function

$$f(a,b,c,d) = a'b'c'd' + a'b'c'd + a'b'ed' + a'b'cd' + a'bcd + abed' + abcd + abcd$$

by karnaugh

Method.

Sol

Step: 1 The binary no/s for the given minterms are 0000, 0001, 0010, 0100, 0101, 0110, 1000, 1001 & 1111

Step: 2

karnaugh table.

ab \ cd	00	01	11	10
00	1	1	0	1
01	1	1	0	1
11	0	0	1	0
10	1	1	0	0

③

$$\begin{array}{r} 0000 \\ 0001 \\ 0100 \\ 0101 \\ \hline abd \end{array}$$

④

$$\begin{array}{r} 1111 \\ \hline abcd \end{array}$$

①

$$\begin{array}{r} 0000 \\ 0001 \\ 1000 \\ 1001 \\ \hline b'c' \end{array}$$

②

$$\begin{array}{r} 0000 \\ 0100 \\ 0010 \\ 0110 \\ \hline a' d' \end{array}$$

Step: 3

Minimum of $f(a,b,c,d)$

$$= a'b'c'd' + a'b'c'd + a'b'ed' + abcd$$